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Monte Carlo simulations of individual variability and their effects on simulated heat stress using thermoregulatory modeling

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This paper addresses a variable-dependence (VD) MC method developed based on a previous attempt (VI-MC method) (J. Therm. Biol. 29 (2004), 515) to be incorporated in a thermoregulatory model. Simulated individuals with anthropometrics by VI- and VD-MC methods for US Army population were compared using principal component analysis and Fisher's exact tests. The results indicated that VD-MC data represented overall body size as the primary component and body shape as the secondary component that were more realistic and similar to the measured US Army data ($p>0.05$) rather than VI-MC data ($p<0.05$). Such differences consequently affected individual thermoregulatory responses to simulated heat stress. The VD-MC method provides a more realistic representation of individual variability and thus underpins more realistic predictions of individual thermoregulatory responses.

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1. Introduction

Thermoregulatory models capable of predicting physiological responses to thermal stress are useful for occupational safety and injury prevention. Predictions of physiological status can be accomplished quickly by simulating the responses to given operational conditions (e.g., work rate and weather), and/or related equipment configuration (e.g., clothing and load weight). Current thermo-physiological simulation models provide reasonably accurate predictions of thermal responses of individuals with specific height, weight, and percent body fat (%BF) (Kraning and Gonzalez, 1997a; Xu et al., 2004). However, for specific populations or groups where individual anthropometric information is not known, the models perform less well due to the lack of information regarding individual variability in the group or population. Over the years, thermoregulatory models have been developed to predict individual variability based on individual anthropometry, basal metabolic rates and %BF (Kraning and Gonzalez, 1997a; Yokota et al., 2005; Zhang et al., 2001). In addition, individual input options such as initial core temperature, heat acclimation status, and factors such as clothing, hydration, circadian rhythm, environment, physical activity that change across time have been incorporated into models (Furlong and Gonzalez, 2003; Kraning and Gonzalez, 1997a; Yokota et al., 2005; Zhang et al., 2001).

Gonzalez (2004) used a Monte Carlo (MC) approach to introduce individual variability associated with physiological responses into a thermoregulatory model. The MC is a random sampling technique utilized in various scientific, economic, and industrial fields. For a thermoregulatory model, MC uses the variable's mean ($\overline{X}$) and standard deviation (SD) for the population and a random number function to generate individual values for the variable. This process is repeated for each variable of interest (e.g., anthropometrics and environment) to obtain random values that are input into the model for predicting thermal responses of individuals (Gonzalez, 2004). The MC method is a useful approach to estimating the fraction of a certain population that will experience thermal-related illness under thermal stressful conditions by randomizing $\overline{X}$ and SD of model variables (e.g., environmental parameters, anthropometric characteristics, and physiological factors). However, this MC approach randomizes each variable independently (Furlong and Gonzalez, 2003; Gonzalez, 2004), although, in reality, anthropometric dimensions are not completely independent (Kroemer et al., 1997; Roebuck, 1995). For instance, tall people tend to be heavier than short people, and a tall person tends to have less %BF than a short person with the same weight. Another MC approach for a thermoregulatory model is to randomize anthropometric inputs utilizing regression equations and percentile methods for creating individual variability (Tikuisis and Keefe, 2005). The regression equations (r=0.34–0.54) to estimate height using weight skew the relationships between anthropometric variables in different occupational populations. In addition, the percentile approach which is relevant to one dimension at a time limits variability of individuals (Zehner et al., 1993).

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In this study, the variable dependent (VD) MC approach was developed, applied, and compared with the traditional variable independent (VI) MC (Gonzalez, 2004) approach. Using these two methods, anthropometric distributions were simulated based on information from US Army anthropometric database (Bathalon et al., 2004), and their effects on a thermoregulatory model were compared. The thermoregulatory model used to simulate physiological responses in this study is based on rational principles of physiology, heat transfer, and thermodynamics (Kraning and Gonzalez, 1997a). The ultimate goal is to improve proper predictions of individual susceptibility to thermal strains when only limited population information is available.

2. Method

The $\overline{X}$ and SD values from the 2004 US Army male anthropometric database (n=1506; height: 177 ± 7(SD) cm; weight: 81 ± 12 kg; BF: 17 ± 6%) (Bathalon et al., 2004) were utilized for both VD- and VI-MC simulations. The correlation (R) matrix constructed based on anthropometric variables from the US Army data is displayed in Table 1. For VD-MC simulations, 100 (generic) points were generated based on the R matrix and scaled to create anthropometric variables using $\overline{X}s$ and SDs of the data. In this way, relationships between and within the variables were maintained. The VD-MC data were compared to VI-MC data, where 100 points were generated independently for each anthropometric variable and then randomly merged to represent the 100 individuals in the group. Fig. 1 summarizes simulation procedures using VD- and VI-MC approaches.

Principal component (PC) analyses were then used in VD-MC, VI-MC, and US Army data to examine the anthropometric distributions. The PC analysis summarizes multivariate data in simple dimensions by transforming data onto linear orthogonal axes (eigenvectors) that maximize data variation (eigenvalues) (Tatsuoka, 1988). Then individual anthropometric variations obtained by a PC analysis were simplified to generate an ellipse encompassing 99% of the population calculated from the centroid. For statistical analysis, outlier boundaries were determined based on the 99% ellipse of the US Army data, and were applied to VD- and VI-MC data. Then, the proportion of outliers in these two datasets was compared with US Army data using Fisher's exact tests (p=0.05).

For thermoregulatory modeling, four primary points on the ellipse of PC dimensions were selected to represent extremes in each population and physiological responses of these individuals to simulated heat stress were compared. The model comprises six compartments (i.e., core, muscle, fat, vascular skin, avascular skin, and central blood volume) to represent the human body. The body temperatures of the various compartments in the model are determined from simultaneous energy balances of each compartment: the active physiological heat control mechanisms (shivering, sweating, skin blood flow) are controlled proportionately by temperature deviations from their set points. The control system of the hypothalamus in the model uses skin, core, and arterial blood temperature. The sweat and blood flow controls are modified by heat production (e.g., metabolic rates and VO_2), dehydration, and heat acclimation. The energy balances determine the rate of temperature changes during the present time (t) and predict the subsequent temperatures in t+1, depending on the rate of change, by stepwise integration.

The model thereby predicts individual heart rates, core and skin temperatures, and sweating rates over time as a function of heat production, anthropometric characteristics (i.e., height, weight, and %BF), thermal aspects of the environment (i.e., ambient temperature, relative humidity, wind speed, and mean radiant temperature) and clothing characteristics (i.e., insulation and vapor permeability), and physiological state (i.e., heat acclimation and hydration). The model was validated under various heat stress conditions including high ambient temperature (e.g., 30–49 °C), strenuous exercise (e.g., 28–72% VO_2max) and various clothing (e.g., light clothing, heavy semi-permeable clothing, and complete encapsulated clothing) (Kraning, 1995; Kraning and Gonzalez, 1997a; Kraning and Gonzalez, 1997b). The detailed mechanism and functions of the model are described elsewhere (Kraning and Gonzalez, 1997a).

The anthropometric values of four extreme individuals identified on the 99% ellipse through PC analyses of each VI- and VD-MC data were applied to the model for comparing physiological responses of these individuals during heat stress. The thermophysiological responses of individuals wearing battle dress uniform (BDU) and body armor, and walking at 1.34 m/s under hot conditions (40 °C, 38% RH) were simulated. The core temperature (T_c) predictions using VD- and VI-MC methods were continued to reach levels of 40 and 42 °C, the core temperatures associated with heat exhaustion and stroke, respectively (McArdle et al., 1996).

Table 1
Correlation coefficient between height (cm), weight (kg) and body fat (%) from 2004 US Army male database.

	Height	Weight	Body fat
Height	1		
Weight	0.50*	1	
Body fat	−0.03	0.70*	1

* Statistical difference at $p < 0.05$.

3. Results

3.1. Anthropometry

Table 2 details the anthropometric distribution patterns based on PC analyses using the VD-MC method. The first principal component (or eigenvalue) (PC1) represents 57% (=100 × 1.70/(1.70+0.84+0.46)) of the total variation and corresponds to all positive loadings of variables in eigenvectors indicating overall size. The second eigenvalue (PC2) represents 28% (=100 × (0.84/(1.70+0.84+0.46))) of the total variation and corresponds with the dichotomous somatic shape in eigenvectors, such as "tall-lean" vs. "short-fat". The third component which explains 15% of the total variability, corresponds to a somatic form such as a football player type (e.g., short-heavy with a low BF%). Fig. 2a summarizes the first two (PC1 and PC2) components listed in Table 2. The primary somatic forms of extreme individuals are identified on the ellipse and defined as "tall-fat (A)", "tall-lean (B)", "short-lean (C)", and "short-fat (D)".

Table 3 presents a summary of PC analysis using the VI-MC method. The anthropometric distributions were different from those found using the VD-MC method. The first eigenvalue, explaining 34% (=100 × 1.03/(1.03+1.01+0.96)) of the total variation, represents the anthropometric patterns observed in the 3rd principal component in VD-MC. The PC2, representing 34% of the total variability (=100/1.01/(1.03+1.01+0.96)), provide the primary information of body weight and fat (e.g., heavy weight with high %BF vs. light weight with low %BF). The third component explains the dichotomous somatic form between tall-heavy weight with low %BF vs. short-light weight with high %BF. Fig. 2b summarizes the first two components corresponding to Table 3. It shows the unrealistic and complicated VI-MC generated somatic forms of extreme individuals on the ellipse

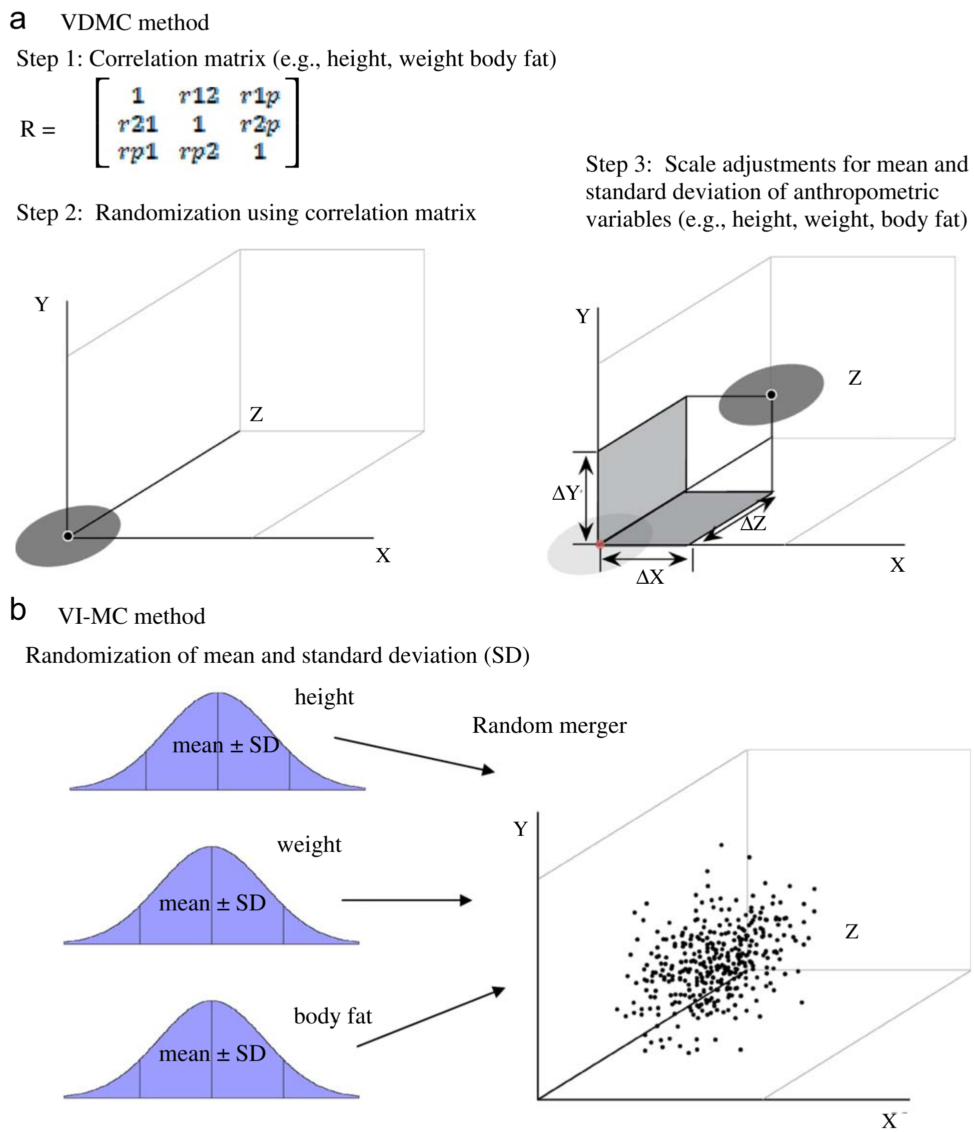

Fig. 1. Variable dependent Monte Carlo (VD-MC) and variable independent Monte Carlo (VI-MC) procedures. (a)VD-MC method. (b)VI-MC method.

Table 2
Principal component analysis summary using variable dependent Monte Carlo (VD-MC) method.

Component	Eigenvalue	Difference	Proportion	Cumulative
1	1.7	0.86	0.57	0.57
2	0.84	0.38	0.28	0.85
3	0.46		0.15	1
Eigenvectors				
Variable	1	2	3	
Height	0.45	0.88	0.17	
Weight	0.65	−0.19	−0.74	
Body fat	0.61	−0.44	0.66	

defined as "tall-thin with high %BF (A)", "fat (B)", "short-heavy with low %BF (C)", and "lean (D)".

Table 4 is the PC analysis summary of US Army data. The overall patterns were similar to the results in the VD-MC method. For quantifying the similarity (or differences) of the PC patterns, outlier boundaries were determined based on the 99% ellipse of the US Army data, and were superimposed on VD- and VI-MC data. Then, the proportion of outliers in these two datasets were compared with US Army data using Fisher's exact tests (p=0.05). Distributions generated by VD-MC captured the pattern of the actual 2004 US Army data, while those simulated by VI-MC differed from both VD-MC and 2004 US Army data. Table 5 summarizes results of Fisher's exact test comparisons between VD-MC (Table 5a), VI-MC (Table 5b) and US Army data. The VD-MC data fit proportionally well with US Army data ($p>0.05$), while the VI-MC data showed more outliers that were statistically different from US Army data ($p<0.05$).

3.2. Thermoregulatory model

Extreme anthropometric somatic forms (A–D) identified by each MC method were converted from PC scores to raw anthropometric scores and applied to the thermoregulatory model. Table 6 shows the summary comparisons of anthropometric values and times to simulated heat exhaustion and stroke by the VD-MC and VI-MC methods. The alphabetical letters (A–D) corresponds to those in the PC dimensions (Fig. 2a and b). The thresholds for heat exhaustion and stroke were based on T_c of 40.0 and 42.0 °C, respectively. The "C" point of the VD-MC method

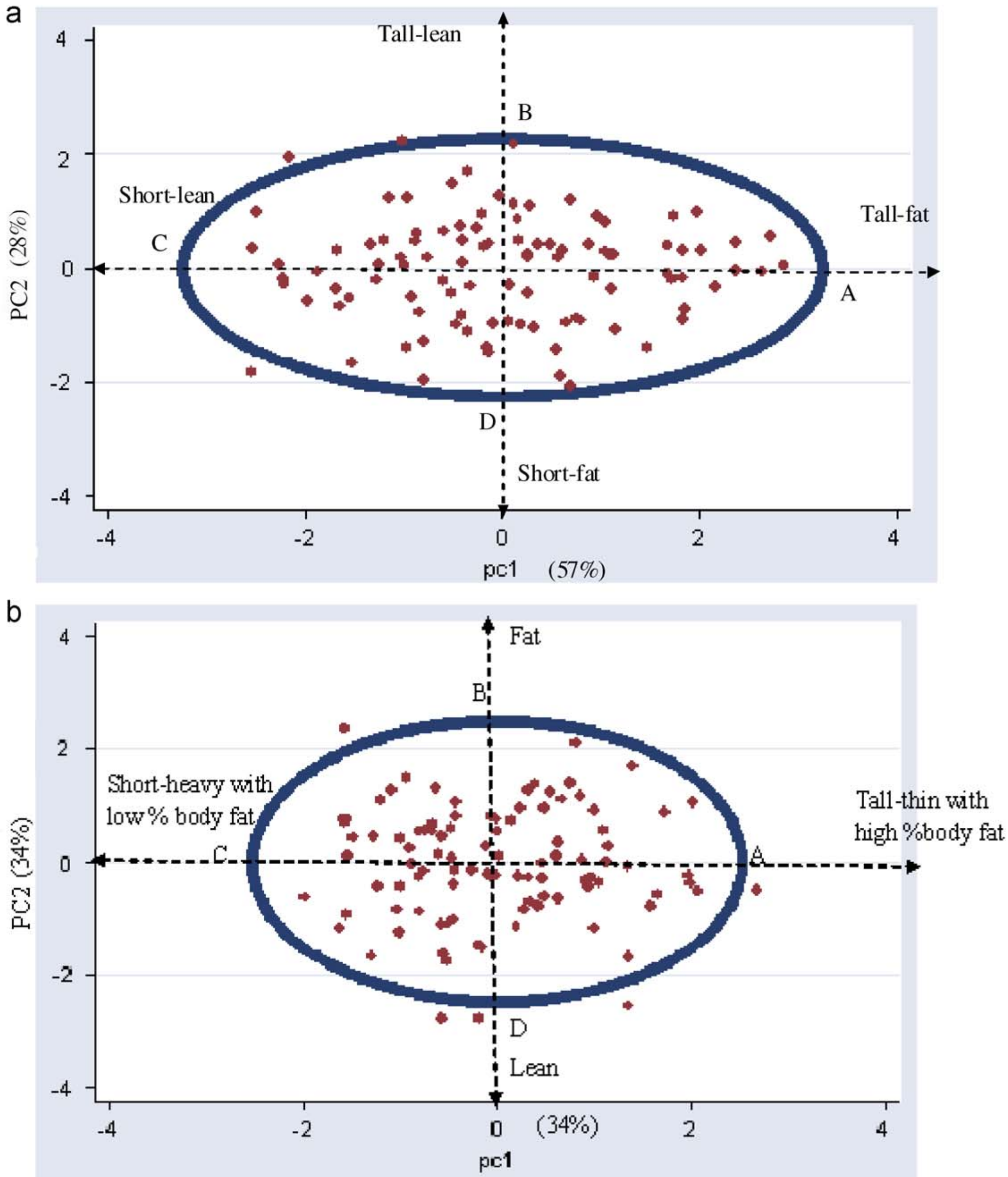

Fig. 2. (a) Principal component (PC) analysis results using the variable dependent Monte Carlo (VD-MC) method (85% of total variability). (b) Principal component (PC) analysis results using the variable independent Monte Carlo (VI-MC) method (68% of total variability).

Table 3
Principal component analysis summary using variable independent Monte Carlo (VI-MC) method.

Component	Eigenvalue	Difference	Proportion	Cumulative
1	1.03	0.02	0.34	0.34
2	1.01	0.05	0.34	0.68
3	0.96		0.32	1
Eigenvectors				
Variable	1	2	3	
height	0.76	0.00	0.66	
Weight	−0.47	0.70	0.54	
Body fat	0.46	0.71	−0.53	

Table 4
Principal component analysis summary of US Army data.

Component	Eigenvalue	Difference	Proportion	Cumulative
1	1.81	0.78	0.6	0.6
2	1.03	0.87	0.34	0.94
3	0.16		0.06	1
Eigenvectors				
Variable	1	2	3	
Height	0.41	0.81	0.43	
Weight	0.71	0.01	−0.7	
Body fat	0.57	−0.59	0.57	

(height: 165 cm weight: 49 kg, body fat: 5%) is more realistic and acceptable than the "C" point of the VI-MC method (height: 161 cm weight: 97 kg, body fat: 9%). Unlike the VI-MC approach, individuals in the VD-MC data indicated variability in tolerance times associated with diverse body size and compositions. Thus, different anthropometric distributions found in PC analysis

Table 5
Fisher's exact test comparisons.

Data	Non-outlier	Outlier	Total
Comparisons between variable dependent Monte Carlo (VD-MC) method and US Army data. (Fisher's exact test p=0.66)			
VD-MC	99	1	100
US Army	1491	15	1506
Total	1590	16	1606
Comparisons between variable independent Monte Carlo (VI-MC) method and US Army data. (Fisher's exact test p=0.000)			
VI-MC	88	12	100
US Army	1491	15	1506
Total	1579	27	1606

Table 6
Summary comparisons of anthropometric values and predicted time to reach heat illness between variable dependent Monte Carlo (VD-MC) method and variable independent Monte Carlo (VI-MC) method.

		Extreme Points			
Methods	Measurements	A	B	C	D
VD-MC	Height (cm)	187	188	165	163
	Weight (kg)	111	79	49	82
	Fat (%)	30	9	5	26
	BSA (m^2)	2.35	2.05	1.52	1.88
	BMI (kg/cm^2)	32	22	18	31
	Time to reach heat exhaustion (min)	83	97	105	85
	Time to reach heat stroke (min)	148	190	231	152
VI-MC	Height (cm)	191	176	161	176
	Weight (kg)	64	105	97	56
	Fat (%)	24	28	9	6
	BSA (m^2)	1.9	2.2	1.99	1.68
	BMI (kg/cm^2)	17	34	38	18
	Time to reach heat exhaustion (min)	106	82	82	106
	Time to reach heat stroke (min)	234	145	145	233

BSA: body surface area; BMI: body mass index.
The thresholds of heat exhaustion and stroke: core temperatures of 40.0 and 42.0 °C, respectively.

affected different physiological responses to simulated heat stress.

Fig. 3 summarizes T_c comparisons among different somatic forms by the VD-MC method. Using a T_c of 40.0 °C as the heat exhaustion threshold, "short-lean" individuals were predicted to perform up to 105 min while "tall-fat" individuals would be expected to perform up to 83 min. Based on the heat stroke threshold, lean individuals were also able to take longer to reach their T_c of 42.0 °C than fat individuals. Overall, "short-lean" individuals, and to a lesser extent "tall-lean" individuals were predicted to be more tolerant of heat stress, maintaining lower T_c than short- and fat-individuals.

4. Discussion

Collecting physiological measures at the population level to assess individual difference of thermal responses is highly impractical because of the time, cost and risk associated with human studies. The incorporation of MC simulation into thermoregulatory mathematical models is an alternative way to simulate the probability that an individual may experience thermal stress under various combinations of environmental and operational conditions. This MC approach is also advantageous when individual anthropometric data is unknown or partially known, or a sample size is too small to predict individual variability in thermal responses. For instance, when examining the effects of newly developed clothing prototypes or protective equipment on human thermal responses to exercise and heat stress, the MC thermoregulatory model estimates individual variability of physiological responses. In addition, planning for rescue missions at sea by the US Coast Guard or by military services may be confounded by ambiguous or completely unknown information. For such situations, the MC thermoregulatory model can estimate a tolerance range of individuals in thermal stressful situations and assess the best and worst scenarios for individual survival, so that rescuers can plan their missions accordingly.

However, the randomization of human dimensions for thermoregulatory model inputs needs to be carefully performed because human dimensions are not completely independent. In this study, individual variability was modeled by the VD-MC method and compared to the traditional VI-MC method. The PC analysis results suggested that different anthropometric distributions were observed between these two methods, resulting in different predicted thermoregulatory responses to simulated heat stress. Anthropometric distributions simulated by VD-MC were primarily represented by overall body size (i.e., tall-heavy with high %BF vs. short-lean with low %BF) and secondarily by the dichotomous shape characteristics (i.e., tall-lean vs. short-fat). Anthropometric distributions simulated by the VI-MC method were explained with unrealistic body shape elements (i.e., PC1: short-lean with heavy body weight vs. tall-fat with light body weight; PC2: high fat and heavy weight vs. lean and light in weight). In general, morphometric/allometric variability in human and animal populations is primarily summarized with size-constrained PC, secondarily with body shape in its subsequent PC (Somers, 1989; Jolicoeur and Mosimann, 1960). Individual anthropometric variability generated by VD-MC was more realistic and statistically similar to those of the measured US Army data than those simulated by the VI-MC method.

The physiological responses to the simulated heat stress in the VD-MC method indicated differences in timing of experiencing thermal-related illness and injuries by somatic forms although the somatic forms are not a direct cause of thermal strain. Lean individuals, having lower %BF and higher body surface per mass, were predicted to maintain lower T_c, and thus take longer to experience heat related illness, than fat individuals. Previous studies of heat stress responses on individuals with different body size, shape and composition (Havenith et al., 1998; Havenith, 2001; Pandolf et al., 1980) supported similar results to this study. Because individual variability represented by somatic forms are different by the VD- and VI-MC methods, physiological responses and tolerance time estimated in the heat stress simulation, i.e., the heat exhaustion time for T_c to reach 40.0 °C, differed between the extreme somatic anthropometric forms simulated by these two methods. Accurate representations of individual variability using the VD-MC approach improve the performance of thermoregulatory models, thereby resulting in more accurate risk/safety assessments.

The VD-MC technique can modify anthropometric characteristics by a population and produce their characteristics more precisely than with the VI-MC approach. For instance, US Army personnel, who are required to meet a strict standard of anthropometry and body composition, are anthropometrically less variable and higher correlations in human dimensions than civilian populations (Roebuck, 1995). Women, in general, with higher %BF, shorter stature, and lighter body weight tend to have different body size and composition than men (Roebuck, 1995). Using a R matrix, $\overline{X}$ and SD of anthropometrics, the VD-MC method can randomize individual variability by maintaining anthropometric characteristics of populations represented by

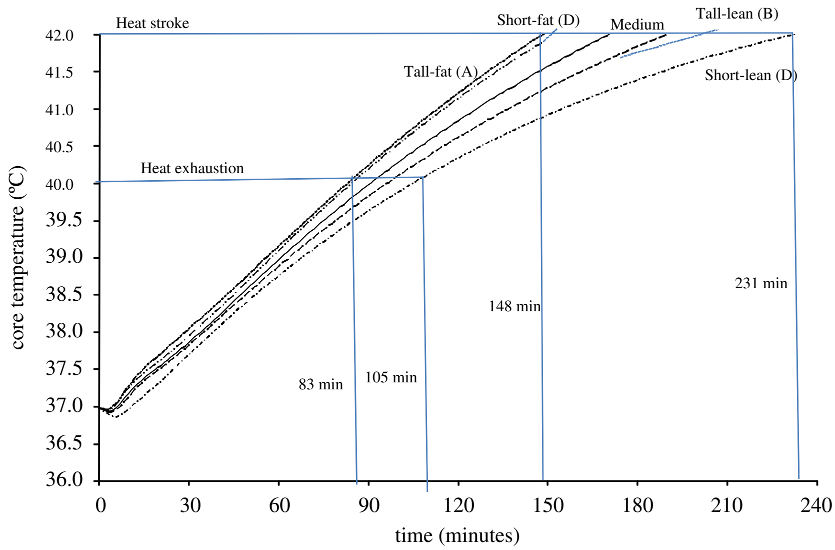

Fig. 3. Core temperature comparisons to simulated heat stress among different somatic forms (A–D) by the variable dependent Monte Carlo (VD-MC) method.

gender, ethnic, or occupational workforces of civilian or military populations. In contrast, VI-MC, which uses a random number generator on an each anthropometric variable and randomly merges anthropometric variables to representatives of individuals, misrepresents the overall anthropometric characteristics of a population. The current VD-MC method provides options for representing either a civilian population or a US Army population, and either pooling or separating the analysis by gender (U.S. Department of Health & Human Services, 2005; Bathalon et al., 2004). Matrices for different populations can be added to the VD-MC method to generate individual variability of anthropometric characteristics that can be used to understand thermo-physiological trends. Furthermore, if necessary, the VD-MC method can incorporate additional values (e.g., VO_2max, age, and circumferences) as inputs into a thermoregulatory mathematical model, and simulate individual variability by retaining characteristics of selected variables.

This study demonstrated the MC method to generate individual variability with realistic human dimensions and its subsequent effects on physiological responses to heat stress simulations. The approach can be further applicable to diverse situations including different occupational missions, the effects of newly developed configurations on thermal responses and performance impairment, predictions of water requirements for logistic supports, as well as strategic actions and planning.

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